

Inequalities

These are like equations, a statement of relation b/w two quantities, but where the $=$ sign is replaced by $<$, $>$, $\leq$, or $\geq$.

Inequalities have solutions just like equations, e.g. $2x+3 > 11$

X	$x=3?$	$\rightarrow$	$9 > 11$	X
X	$x=4?$	$\rightarrow$	$11 > 11$	
✓	$x=5?$	$\rightarrow$	$13 > 11$	

Intervals

Notation	Inequality	Graph
(1) (a, b)	$a < x < b$	
(2) $[a, b]$	$a \leq x \leq b$	
(3) $[a, b)$	$a \leq x < b$	
(4) $(a, b]$	$a < x \leq b$	
(5) (a, ∞)	$x > a$	
(6) $[a, \infty)$	$x \geq a$	
(7) $(-\infty, b)$	$x < b$	
(8) $(-\infty, b]$	$x \leq b$	
(9) $(-\infty, \infty)$	$-\infty < x < \infty$	

e.g. $4x - 3 < 2x + 5$

Method 1

$$4x - 2x < 5 + 3$$

$$\Rightarrow 2x < 8$$

$$\Rightarrow x < 4$$

$$(-\infty, 4)$$

Method 2

$$-3 - 5 < 2x - 4x$$

$$\Rightarrow -8 < -2x$$

$$\Rightarrow 4 > x$$

$$(-\infty, 4)$$

mult.
ordiv.
by -
flips
ineq.

continued inequality, e.g. $-6 < 2x - 4 \leq 2$
(conjunction)

$$-6 < 2x - 4 \quad \& \quad 2x - 4 \leq 2$$

$$-6 < 2(\otimes) - 4 \leq 2$$

$$-6 + 4 < 2x \leq 2 + 4$$

$$-2 < 2x \leq 6$$

$$-2 < x \leq \frac{6}{2}$$

$$-1 < x \leq 3$$

$$-1 < x$$

$$(-1, \infty)$$

$$x \leq 3$$

$$(-\infty, 3]$$

$$(-1, 3] = (-1, \infty) \cap (-\infty, 3]$$

rational inequality, e.g. $\frac{-2}{4-3x} > 0$

$$\frac{4-3x}{-2} > 0$$

$$x \neq \frac{4}{3}$$

$\Downarrow$

$$-2 + \frac{3}{2}x > 0$$

$\Downarrow$

$$\frac{3x}{2} > 2$$

$$x > \frac{4}{3}$$

$$\frac{4-3x}{-2} = \frac{1}{\left(\frac{-2}{4-3x}\right)}$$

$$4-3x < 0$$

$$x > \frac{4}{3}$$

$$\left(\frac{4}{3}, \infty\right)$$

absolute-value inequalities, e.g. $|x| < 3$

$$|x| < 3 \iff -3 < x < 3$$

$$\iff x \in (-3, 3)$$

$$\frac{a}{b} > 0$$

$$a > 0 \text{ \& } b > 0$$

OR

$$a < 0 \text{ \& } b < 0$$

$$\frac{-2}{b} > 0$$

~~$$-2 > 0 \text{ \& } b > 0$$~~

OR

$$-2 < 0 \text{ \& } b < 0$$

$$\frac{-2}{4-3x} > 0 \Leftrightarrow \frac{-2}{4-3x} \text{ is positive}$$

$$\Leftrightarrow 4-3x \text{ is negative}$$

$$\Leftrightarrow 4-3x < 0$$

e.g. $|2x+3| > 9$

$2x+3$ is "distance" ~~at least~~ more
than 9 from 0
(disjunction)

$$2x+3 < -9 \quad \text{OR} \quad 2x+3 > 9$$

$$\downarrow$$

$$2x < -12$$

$$\downarrow$$

$$x < -6$$

$$(-\infty, -6)$$

$$\downarrow$$

$$2x+3 > 9$$

$$\downarrow$$

$$2x > 6$$

$$\downarrow$$

$$x > 3$$

$$(3, \infty)$$

Final answer: $(-\infty, -6) \cup (3, \infty)$

e.g.

$$\frac{2}{|2x+3|} \geq 5$$

$$2x+3 \neq 0$$

$$x \neq -\frac{3}{2}$$

If we're in the domain,

$$2x+3 \neq 0 \implies |2x+3| > 0$$

↓
positive

multiply both sides by $|2x+3|$

$$2 \geq 5 |2x+3| \quad \frac{2}{5} \geq |2x+3|$$

$$|2x+3| \leq \frac{2}{5}$$

$$-\frac{2}{5} \leq 2x+3 \leq \frac{2}{5}$$

$$-\frac{17}{5} \leq 2x \leq -\frac{13}{5}$$

$$-\frac{17}{10} \leq x \leq -\frac{13}{10}$$

$$\left[-\frac{17}{10}, -\frac{13}{10}\right]$$